



ELEC 475

Lab 4 – Semantic Segmentation & Knowledge Distillation

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Pre-Lab

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1. Introduction

The goal of this lab is to explore the fundamentals of semantic segmentation and apply a knowledge distillation technique. Semantic segmentation involves generating a pixel-wise segmentation mask for an image, assigning classification labels to each pixel. Knowledge distillation, as introduced by Hinton et al., transfers knowledge from a larger model to a smaller one, enabling deployment on devices with limited computational resources (e.g., phones or smartwatches) while maintaining reasonable accuracy and processing speed. In this lab, you will design, implement, train, and evaluate a compact model that distills knowledge from an FCN-ResNet50 pre-trained on the PASCAL VOC 2012 Dataset.

You may use any large language model (e.g., ChatGPT, Claude) for this assignment. When prompting, be explicit and precise about what you want. Always test and fact-check every piece of code the model generates — do not accept code outputs at face value. Verify imports, function names, arguments, and tensor shapes, etc. - and of course make sure that it runs correctly!

2. Task

Your pre-lab task comprises the following single step:

2.1 Step One – Test Pretrained FCN-ResNet50 on PASCAL VOC 2012 Dataset

Create a test script that loads PyTorch's pretrained FCN-ResNet50 model, downloads the Pytorch segmentation version of PASCAL VOC 2012 dataset, and evaluates the model using the mean Intersection over Union (mIoU) metric.

Hint 1: Make the most of PyTorch's official documentation and tutorials—they provide valuable insights into the task, the mIoU metric, and can help streamline utility function setup, saving time and enhancing your understanding.

Hint 2: You may find it useful to prompt your LLM to draft a minimal example of loading and evaluating PyTorch's pretrained FCN-ResNet50 on the PASCAL VOC 2012 dataset. Ask for clear code showing dataset preparation, model evaluation, and mIoU calculation. Examine every import, tensor shape, and function call before running—treat the response as the starting point, not as the final code.